Question ID 55ea82f3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 55ea82f3

A team hosting an event to raise money for new uniforms plans to sell at least 140 tickets before this event and at least 220 tickets during this event to raise a total of at least $\$5{,}820$ from all tickets sold. The price of a ticket during this event is $\$3$ less than the price of a ticket before this event. Which inequality represents this situation, where x is the price, in dollars, of a ticket sold during this event?

A. $140(x+3)+220x \leq 5{,}820$

B. $140(x+3)+220x \geq 5{,}820$

C. $140(x-3)+220x \leq 5{,}820$

D. $140(x-3)+220x \geq 5{,}820$

ID: 55ea82f3 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that a team plans to sell at least 140 tickets before an event and at least 220 tickets during the event to raise a total of at least $\$5{,}820$ from all tickets sold. It's also given that the price of a ticket during the event is $\$3$ less than the price of a ticket before the event and that x represents the price, in dollars, of a ticket sold during the event. It follows that $x+3$ represents the price, in dollars, of a ticket sold before the event. The expression $140(x+3)$ represents the planned revenue, in dollars, from the tickets sold before the event, and the expression $220x$ represents the planned revenue, in dollars, from the tickets sold during the event. Thus, the expression $140(x+3)+220x$ represents the planned revenue, in dollars, from all tickets sold. Since the team plans to raise a total of at least $\$5{,}820$ from all tickets sold, the total revenue must be at least $\$5{,}820$. Therefore, the inequality $140(x+3)+220x \geq 5{,}820$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which the team raises a total of at most $\$5{,}820$ from all tickets sold.

Choice C is incorrect. This inequality represents a situation in which the price of a ticket before the event is $\$3$ less, rather than $\$3$ more, than the price of a ticket during the event and the team raises a total of at most $\$5{,}820$ from all tickets sold.

Choice D is incorrect. This inequality represents a situation in which the price of a ticket before the event is $\$3$ less, rather than $\$3$ more, than the price of a ticket during the event.

Question Difficulty:
Hard